



25 Most Important SQL Queries to Crack Any In interview 2025

Table Structure Used in Examples

Basic SQL Queries

JOINS

Filtering and Sorting

Aggregations & Subqueries

Nested Queries / Correlated Subqueries

Window Functions (Advanced)

Set Operators

String & Date Functions

Practice-Style Queries

1. **Get all employee records.**

```
SELECT * FROM Employees;
```

2. **Get names of employees from the IT department.**

```
SELECT Name FROM Employees WHERE Department = 'IT';
```

3. **Get employees with salary greater than 55000.**

```
SELECT * FROM Employees WHERE Salary > 55000;
```

4. **Count number of employees in each department.**

```
SELECT Department, COUNT(*) FROM Employees GROUP BY Department;
```

5. **Get maximum salary in each department.**

```
SELECT Department, MAX(Salary) FROM Employees GROUP BY Department;
```

6. **Get employee names along with their department names.**

```
SELECT E.Name, D.DeptName FROM Employees E  
JOIN Departments D ON E.Department = D.DeptName;
```

7. **Find employees with no valid department in the Departments table.**

```
SELECT E.Name FROM Employees E LEFT JOIN Departments D ON E
```

8. **List top 3 highest-paid employees.**

```
SELECT * FROM Employees ORDER BY Salary DESC LIMIT 3;
```

9. **Get employees with salary between 50000 and 60000.**
SELECT * FROM Employees WHERE Salary BETWEEN 50000 AND 60000;
10. **List departments with more than one employee.**
SELECT Department, COUNT(*) FROM Employees GROUP BY Department HAVING COUNT(*) > 1;
11. **Get the average salary of all employees.**
SELECT AVG(Salary) FROM Employees;
12. **Find employees earning more than the average salary.**
SELECT * FROM Employees WHERE Salary > (SELECT AVG(Salary) FROM Employees);
13. **Get the department with the highest total salary.**
SELECT Department FROM Employees GROUP BY Department ORDER BY SUM(Salary) DESC LIMIT 1;
14. **Find the highest-paid employee in each department.**
**SELECT * FROM Employees E1 WHERE Salary = (
SELECT MAX(Salary) FROM Employees E2 WHERE E1.Department = E2.Department);**
15. **Find employees who earn the highest salary overall.**
SELECT * FROM Employees WHERE Salary = (SELECT MAX(Salary) FROM Employees);
16. **Rank employees by salary within departments.**
**SELECT Name, Department, Salary,
RANK() OVER (PARTITION BY Department ORDER BY Salary DESC) AS SalaryRank FROM Employees;**

17. **Show cumulative salary by department.**
SELECT Name, Department, Salary,
SUM(Salary) OVER (PARTITION BY Department ORDER BY Salary) AS CumSalary FROM Employees;
18. **Get departments listed in Employees but not in Departments.**
SELECT DISTINCT Department FROM Employees WHERE Department NOT IN (SELECT DeptName FROM Departments);
19. **Get employees whose names start with 'A'.**
SELECT * FROM Employees WHERE Name LIKE 'A%';
20. **Format salary with a currency symbol (MySQL).**
SELECT Name, CONCAT('\$', FORMAT(Salary, 2)) AS Salary FROM Employees;
21. **Find the second highest salary.**
SELECT MAX(Salary) FROM Employees WHERE Salary < (SELECT MAX(Salary) FROM Employees);
22. **Find duplicate department entries in Employees.**
SELECT Department, COUNT(*) FROM Employees GROUP BY Department HAVING COUNT(*) > 1;
23. **Order employees by department and then salary descending.**
SELECT * FROM Employees ORDER BY Department, Salary DESC;
24. **List all unique departments across Employees and Departments.**
SELECT Department FROM Employees UNION SELECT DeptName FROM Departments;
25. **List departments where no employee works.**
SELECT DeptName FROM Departments WHERE DeptName NOT IN (SELECT Department FROM Employees);